NEET UG BIOLOGY • UNIT: REPRODUCTION

Sexual Reproduction in Flowering Plants

A comprehensive, self-contained tutorial covering angiosperm floral morphology, gametogenesis, pollination biology, double fertilisation, embryogeny, seed and fruit formation, and apomixis.

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1. Learning Objectives & Historical Overview

Biology is the story of life on Earth. While individual organisms inevitably die, species endure over millions of years through **reproduction**. Sexual reproduction creates new genetic variants, enhancing survival advantage in changing environments. All flowering plants (angiosperms) exhibit sexual reproduction, culminating in the formation of fruits and seeds.

Historical Spotlight: Professor Panchanan Maheshwari (1904–1966)

- Born in November 1904 in Jaipur (Rajasthan), Panchanan Maheshwari became one of the most distinguished botanists in India and the world.
- He obtained his D.Sc. in Allahabad, where he was inspired by Dr. W. Dudgeon, an American missionary teacher, to pursue botany and plant morphology.
- **Key Scientific Contributions:**
  - Popularised the use of **embryological characters in taxonomy**.
  - Established the Department of Botany at the University of Delhi as a premier global centre for research in **embryology and tissue culture**.
  - Emphasised the initiation of work on the **artificial culture of immature embryos**.
  - Achieved worldwide acclaim for his pioneering work on **test-tube fertilisation** and **intra-ovarian pollination**.
- He was honoured with the Fellowship of the Royal Society of London (FRS) and the Indian National Science Academy, and led the publication of the very first NCERT Biology textbooks for Higher Secondary Schools in 1964.

Core NEET UG Learning Competencies

- Identify the microscopic anatomy of the anther, microsporangium wall layers, and male gametophyte (pollen grain).
- Trace megasporogenesis and the monosporic development of the 7-celled, 8-nucleate female gametophyte (embryo sac).
- Compare autogamy, geitonogamy, and xenogamy, alongside abiotic and biotic pollination adaptations and outbreeding devices.
- Explain pollen-pistil interaction, artificial hybridisation (emasculation and bagging), and double fertilisation (syngamy + triple fusion).
- Differentiate post-fertilisation structures: nuclear vs. cellular endosperm, dicot vs. monocot embryos, albuminous vs. non-albuminous seeds, true vs. false vs. parthenocarpic fruits, apomixis, and polyembryony.

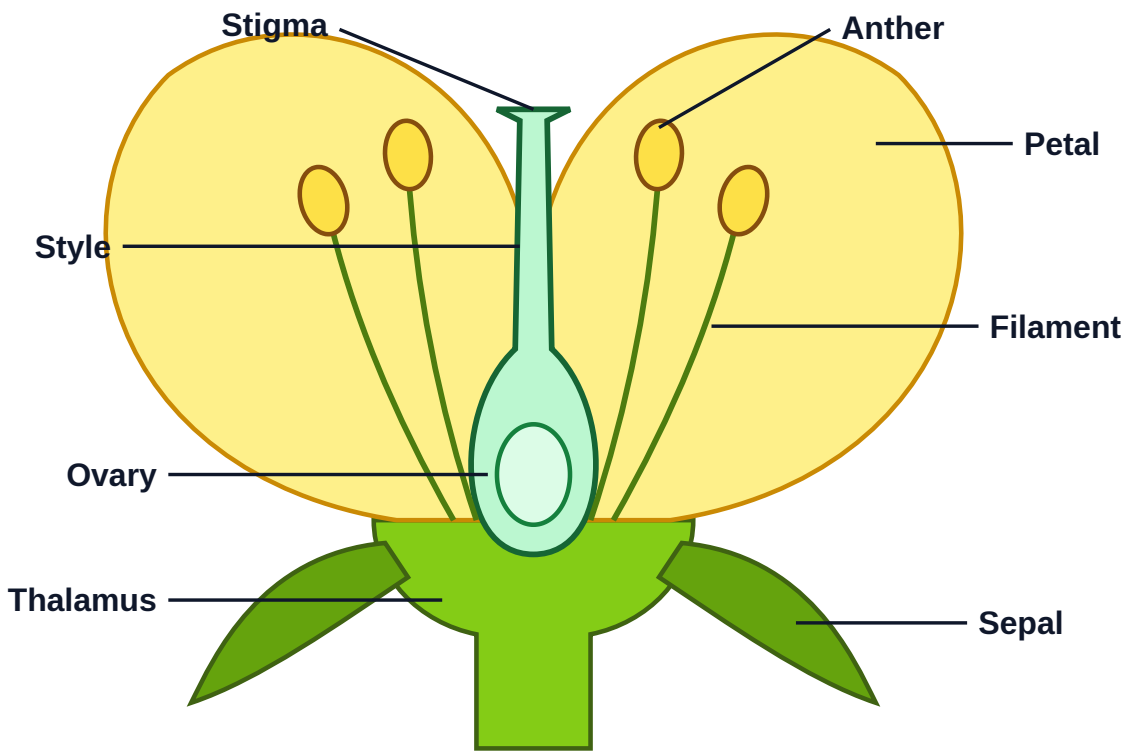
2. Flower — A Fascinating Organ of Angiosperms

Human beings have shared an intimate relationship with flowers since antiquity, using them as objects of aesthetic, ornamental, social, religious, and cultural value. The cultivation, management, and marketing of flowers is known as **floriculture**.

To a biologist, flowers are morphological and embryological marvels that serve as the sites of sexual reproduction. Much before a flower is visible on a plant, the physiological decision to flower has already occurred. Hormonal and structural changes initiate the differentiation and development of the **floral primordium**. Inflorescences form, bearing floral buds that mature into flowers.

Within the flower, the male and female reproductive structures differentiate:

- **Androecium:** A whorl of stamens representing the male reproductive organ.
- **Gynoecium:** A single pistil or whorl of pistils representing the female reproductive organ.



Tutorial Diagram 1: Longitudinal section (L.S.) of a typical angiosperm flower showing the floral whorls.

3. Pre-Fertilisation: Stamen, Microsporangium & Pollen Grain

3.1 Structure of a Typical Stamen and Anther

A typical stamen consists of two main parts:

1. **Filament:** A long and slender stalk. The **proximal end** of the filament is attached to the **thalamus** or the **petal** of the flower (epipetalous condition). The number and length of stamens vary across species.
2. **Anther:** The terminal, generally **bilobed** structure. Each lobe possesses two theca, meaning a typical angiosperm anther is **dithecous**. A longitudinal groove often runs lengthwise, separating the theca.

In transverse section, the anther is a **four-sided (tetragonal) structure** consisting of **four microsporangia** located at the corners—two in each lobe (**tetrasporangiate**). As the microsporangia develop further, they become **pollen sacs** that extend longitudinally through the entire length of the anther and become packed with pollen grains.

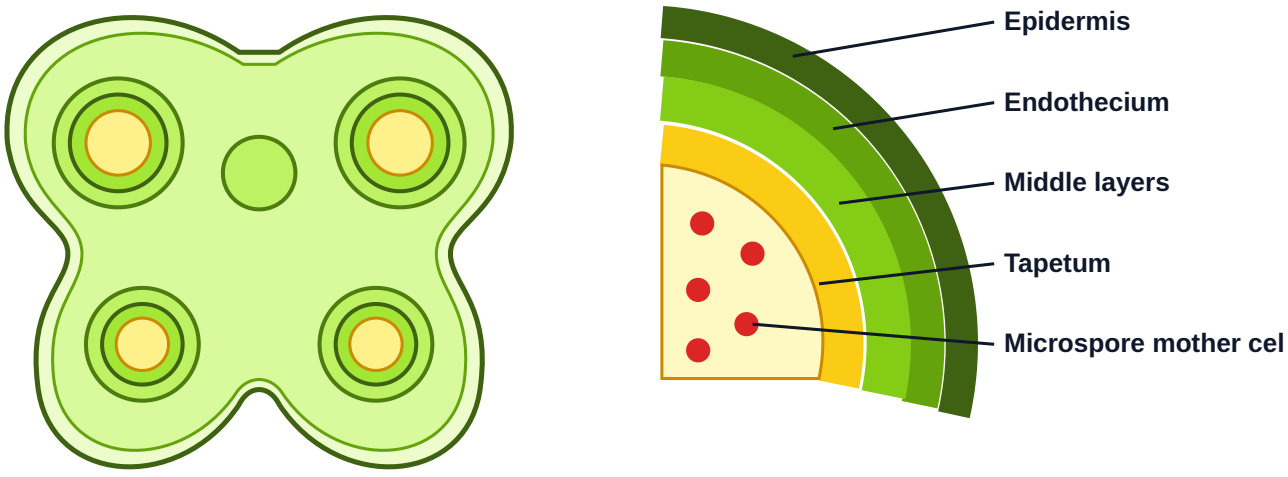
3.2 Structure of the Microsporangium

In a transverse section, a typical young microsporangium appears nearly circular in outline and is surrounded by **four wall layers** (from outside to inside):

1. **Epidermis:** Outermost single protective layer.
2. **Endothecium:** Second layer, featuring fibrous thickenings that assist in anther dehiscence.
3. **Middle layers:** 1 to 3 layers of cells internal to the endothecium; degenerate as the anther matures.
4. **Tapetum:** The innermost wall layer.

Functions of Anther Wall Layers & Tapetum

- **Outer three layers (Epidermis, Endothecium, Middle layers):** Perform the function of **protection** and help in the **dehiscence of the anther** to release pollen.
- **Tapetum (Innermost layer): Nourishes the developing pollen grains** and contributes to the formation of the sporopollenin-rich outer pollen wall (exine). Cells of the tapetum possess **dense cytoplasm** and generally have **more than one nucleus** (become binucleate or multinucleate due to endomitosis or free nuclear division without cytokinesis).
- **Sporogenous tissue:** When the anther is young, a group of compactly arranged, homogeneous diploid cells occupies the centre of each microsporangium.



(a) T.S. of Young Anther **(b) Enlarged Microsporangium Wall Layers**

Tutorial Diagram 2: (a) Transverse section of a young tetrasporangiate anther; (b) Enlarged view of microsporangium wall layers from epidermis to tapetum surrounding microspore mother cells.

3.3 Microsporogenesis

As the anther develops, the cells of the sporogenous tissue undergo meiotic divisions to form **microspore tetrads**. Because each cell of the sporogenous tissue is capable of giving rise to a microspore tetrad, each cell is a potential **pollen mother cell (PMC)** or **microspore mother cell** $(2n)$.

Definition: The process of formation of haploid microspores (n) from a diploid pollen mother cell (PMC, $2n$) through meiosis is called **microsporogenesis**.

- Newly formed microspores are arranged in a cluster of four haploid cells called the **microspore tetrad** (each cell has ploidy n).

- As the anthers mature and **dehydrate**, the microspores dissociate from one another and develop into individual **pollen grains**.
- Inside each microsporangium, several thousands of pollen grains are formed and released upon anther dehiscence.

3.4 Pollen Grain (Male Gametophyte)

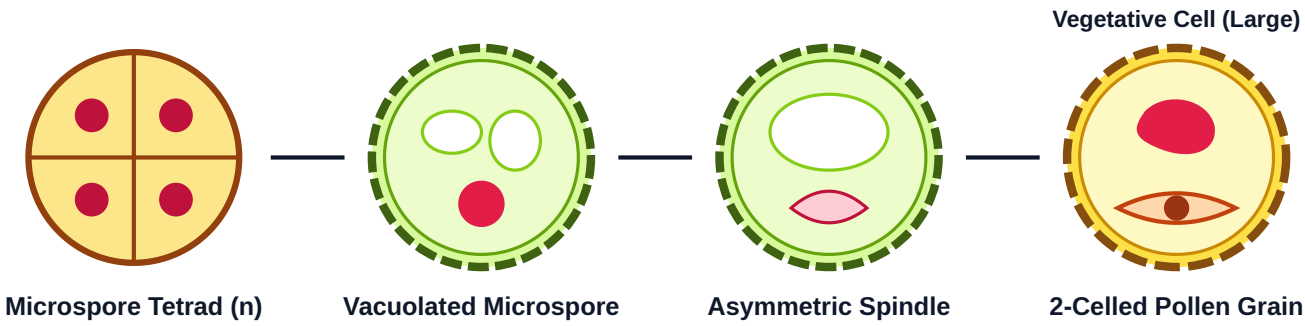
The pollen grains represent the **male gametophytes**. They are generally spherical, measuring about $25–50\ \mu m$ (micrometers) in diameter, and possess a prominent two-layered wall (sporoderm):

Feature	Exine (Outer Wall)	Intine (Inner Wall)
Chemical Composition	Composed of **sporopollenin**, one of the most resistant organic materials known.	Composed of **cellulose and pectin**.
Structural Continuity	Hard, discontinuous layer; features prominent apertures called **germ pores** where sporopollenin is absent.	Thin and continuous layer surrounding the plasma membrane of the cytoplasm.
Biological Significance	Withstands high temperatures and strong acids and alkalis; **no enzyme that degrades sporopollenin is known so far**. Enables pollen grains to be well-preserved as **fossils** and exhibits species-specific sculpturing patterns.	Emerges through one of the germ pores during pollen germination to form the **pollen tube**.

Maturation of Microspore into a Pollen Grain

As a haploid microspore matures into a pollen grain, vacuoles develop and the nucleus undergoes an asymmetric mitotic division via an **asymmetric spindle**, producing two unequal cells:

1. **Vegetative cell (Tube cell):** Bigger, contains abundant food reserve, and possesses a **large, irregularly shaped nucleus**.
2. **Generative cell:** Small, **spindle-shaped** cell with dense cytoplasm and a nucleus; it **floats in the cytoplasm of the vegetative cell**.



Tutorial Diagram 3: Stages of a microspore maturing into a 2-celled pollen grain (vegetative cell and spindle-shaped generative cell).

- **2-Celled vs. 3-Celled Shedding Stage:**
  - In **over 60% of angiosperms**, pollen grains are shed at the **2-celled stage** (1 vegetative cell + 1 generative cell).
  - In the remaining species (about 40%), the generative cell divides mitotically to form **two male gametes** before the pollen grains are shed, resulting in shedding at the **3-celled stage** (1 vegetative cell + 2 male gametes).

- **Pollen Allergy:** Pollen grains of many species cause severe allergies and bronchial afflictions, often leading to chronic respiratory disorders such as **asthma and bronchitis**. *Parthenium* (carrot grass), which entered India as a contaminant with imported wheat, has become ubiquitous and is a major cause of pollen allergy.
- **Pollen as Nutrient Supplements:** Pollen grains are rich in nutrients. In western countries, pollen tablets and syrups are widely marketed as food supplements claimed to enhance the performance of **athletes and race horses**.
- **Pollen Viability and Cryopreservation:**
  - Once shed, pollen grains must land on a compatible stigma before losing viability to bring about fertilisation. Viability depends on prevailing **temperature and humidity**.
  - In cereals such as **rice and wheat**, pollen grains lose viability within **30 minutes** of release.
  - In some members of **Rosaceae, Leguminosae, and Solanaceae**, pollen grains maintain viability for **months**.
  - **Pollen Banks:** Pollen grains of many species can be stored for years in **liquid nitrogen at** $-196\,^{\circ}\mathrm{C}$ (cryopreservation) for use in crop breeding programmes, similar to seed banks or semen storage.

4. Pre-Fertilisation: Pistil, Megasporangium (Ovule) & Embryo Sac

4.1 The Gynoecium and Pistil

The gynoecium represents the female reproductive part of the flower:

- **Monocarpellary:** Consists of a single pistil.
- **Multicarpellary:** Consists of more than one pistil.
  - **Syncarpous:** Pistils are fused together (e.g., *Papaver* / poppy, *Hibiscus*).
  - **Apocarpous:** Pistils are free from one another (e.g., *Michelia*).

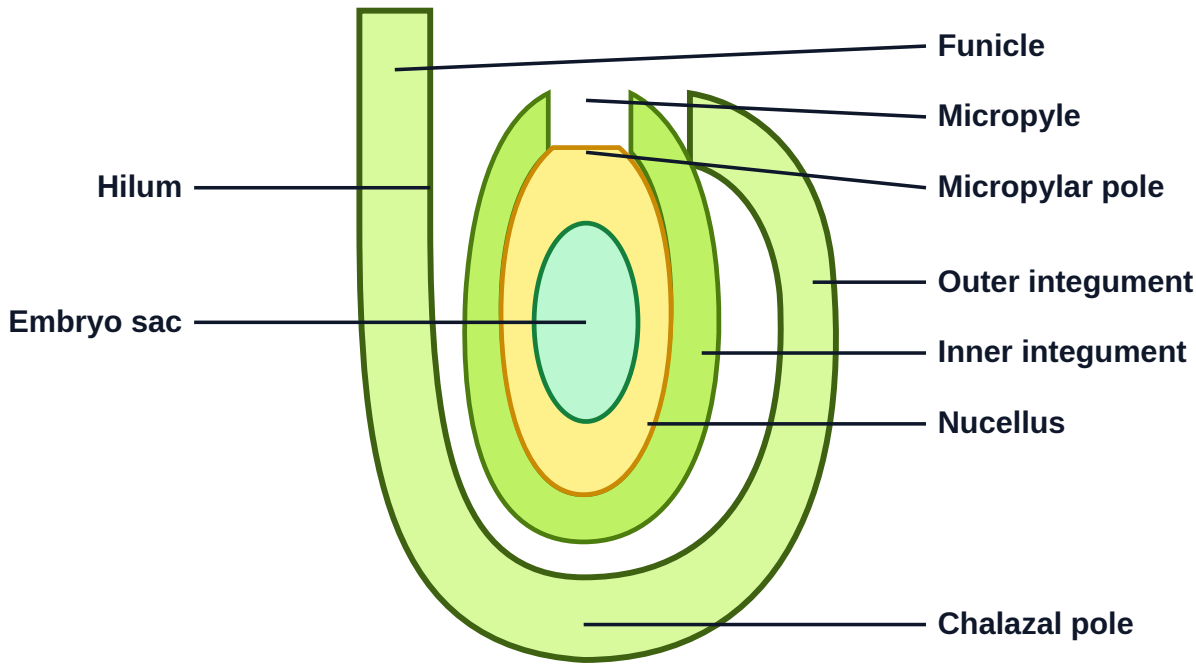
Each pistil consists of three parts: the **stigma** (landing platform for pollen grains), the **style** (elongated slender part beneath the stigma), and the basal bulged **ovary**. Inside the ovary is the **ovarian cavity (locule)** containing the **placenta**, from which arise the megasporangia, commonly called **ovules**.

- **Single ovule per ovary:** Wheat, paddy (rice), mango.
- **Many ovules per ovary:** Papaya, watermelon, orchids.

4.2 Structure of the Megasporangium (Anatropous Ovule)

A typical angiosperm ovule is an **anatropous (inverted) ovule** with the following distinct parts:

- **Funicle:** The stalk that attaches the ovule to the placenta.
- **Hilum:** The region where the body of the ovule fuses with the funicle (junction between ovule and funicle).
- **Integuments:** One or two protective envelopes (outer and inner integuments) that encircle the nucellus except at the tip.
- **Micropyle:** A small opening organised at the tip where integuments are absent (marks the **micropylar pole**).
- **Chalaza:** Located opposite the micropylar end, representing the **basal part** of the ovule (**chalazal pole**).
- **Nucellus:** A mass of parenchymatous diploid ($2n$) cells enclosed within the integuments, containing **abundant reserve food materials**.
- **Embryo sac (Female gametophyte):** Located within the nucellus; an ovule generally has a single embryo sac formed from a functional haploid megaspore.



Tutorial Diagram 4: Diagrammatic view of a typical angiosperm anatropous ovule (megasporangium).

4.3 Megasporogenesis and Monosporic Embryo Sac Development

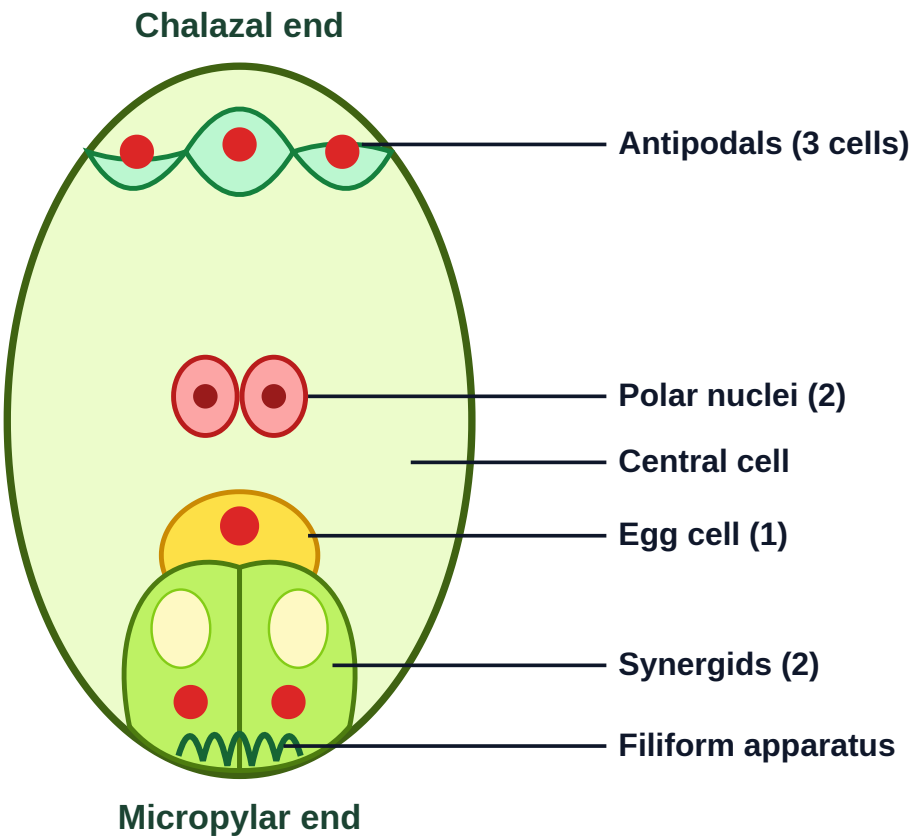
Megasporogenesis is the process of formation of haploid megaspores from the diploid **megaspore mother cell (MMC)**.

1. Ovules generally differentiate a **single MMC** in the **micropylar region of the nucellus**. The MMC is a large diploid ($2n$) cell with dense cytoplasm and a prominent nucleus.
2. The MMC undergoes **meiotic division** (Meiosis I forms a megaspore dyad; Meiosis II forms a linear **tetrad of four haploid megaspores**).
3. **Monosporic Development:** In a majority of flowering plants, **three megaspores degenerate** and only **one megaspore remains functional** (typically the chalazal one). The development of the female gametophyte (embryo sac) from a single functional megaspore is termed **monosporic development**.

Sequential Free-Nuclear Divisions and Organisation of the Embryo Sac

- The nucleus of the functional haploid megaspore divides mitotically into two nuclei that move to opposite poles, forming the **2-nucleate embryo sac**.
- Two more sequential mitotic nuclear divisions produce the **4-nucleate** and subsequently the **8-nucleate stages** of the embryo sac.
- **Important Note:** These three mitotic divisions are strictly **free-nuclear** (nuclear divisions are not immediately followed by cell wall formation).
- After the 8-nucleate stage, cell walls are laid down around six of the eight nuclei, organising the typical mature female gametophyte:
  - **Egg Apparatus (at the Micropylar End — 3 cells):** Consists of **two synergids** and **one egg cell**. The synergids possess special cellular thickenings at their micropylar tip called the **filiform apparatus**, which guides the pollen tube into the synergid.
  - **Antipodals (at the Chalazal End — 3 cells):** Three haploid vegetative cells at the basal chalazal pole.
  - **Central Cell (1 large cell with 2 nuclei):** Situated below the egg apparatus; contains **two haploid polar nuclei** ($n + n$).

Therefore, a typical angiosperm embryo sac at maturity is **8-nucleate and 7-celled**.



Tutorial Diagram 5: Mature 7-celled, 8-nucleate angiosperm embryo sac (female gametophyte). Note that NCERT diagrams depict both orientations (micropylar end at top or bottom); always orient by the filiform apparatus at the micropylar end and antipodals at the chalazal end.

Ploidy Reference Table of Gametophytic & Sporophytic Structures

Structure / Cell	Ploidy Level	Generation & Origin
Nucellus, Integuments, Funicle	Diploid ($2n$)	Sporophytic maternal tissue
Pollen Mother Cell (PMC) / Microspore Mother Cell	Diploid ($2n$)	Sporophytic cell undergoing meiosis
Megaspore Mother Cell (MMC)	Diploid ($2n$)	Single sporophytic cell in micropylar nucellus undergoing meiosis
Microspore / Pollen Grain (Vegetative & Generative cells)	Haploid (n)	Male gametophyte formed after meiosis
Functional Megaspore	Haploid (n)	First cell of female gametophyte formed after meiosis
Synergids, Egg cell, Antipodals, Polar nuclei	Haploid (n each)	Cells/nuclei of the mature female gametophyte (embryo sac)

5. Pollination: Types, Agents, Outbreeding Devices & Pollen-Pistil Interaction

Because both male and female gametes in flowering plants are **non-motile**, they must be brought together for fertilisation. **Pollination** is the transfer of pollen grains (shed from the anther) to the stigma of a pistil.

5.1 Kinds of Pollination (Based on Source of Pollen)

Type of Pollination	Definition	Genetic & Ecological Features
1. Autogamy (Self-pollination)	Transfer of pollen grains from the anther to the stigma of the **same flower**.	Requires synchrony in pollen release and stigma receptivity, and close proximity between anther and stigma. Rare in normal open flowers.
2. Geitonogamy	Transfer of pollen grains from the anther to the stigma of **another flower of the same plant**.	**Functionally cross-pollination** (requires a pollinating agent), but **genetically similar to autogamy** because pollen comes from the same plant.
3. Xenogamy	Transfer of pollen grains from the anther to the stigma of a **different plant**.	The **only type of pollination** that brings **genetically different** types of pollen grains to the stigma.

Chasmogamous vs. Cleistogamous Flowers

Plants such as ***Viola* (common pansy), *Oxalis*, and *Commelina*** produce two types of flowers on the same plant:

- **Chasmogamous flowers:** Open flowers with exposed anthers and stigma, similar to flowers of other species.
- **Cleistogamous flowers:** Flowers that **do not open at all**. Anthers and stigma lie close to each other inside the closed bud. Cleistogamous flowers are **invariably autogamous** (zero chance of cross-pollen landing on the stigma) and guarantee **assured seed-set even in the absence of pollinators**. However, continuous self-pollination leads to inbreeding depression and lack of genetic variation.

5.2 Agents of Pollination

Plants use two **abiotic agents (wind and water)** and one **biotic agent (animals)**. The vast majority of plants use biotic agents. Because pollen contact with the stigma in wind and water pollination is a chance event, abiotic-pollinated flowers produce an **enormous amount of pollen** compared to the number of ovules available, and their flowers are **not very colourful and do not produce nectar**.

A. Wind Pollination (Anemophily)

- Most common amongst abiotic pollinations; very common in **grasses**.
- **Floral Adaptations:**
  - Pollen grains are **light and non-sticky** (easily transported in wind currents).
  - Stamens are **well-exposed** for easy dispersal into wind currents.
  - Stigma is **large and often feathery** to trap airborne pollen grains.
  - Flowers typically have a **single ovule in each ovary** and numerous flowers packed into an inflorescence.
- **Classic Example: Corn cob (Maize)** — the tassels/ears waving in the wind are nothing but the elongated **stigma and style** designed to trap airborne pollen grains.

B. Water Pollination (Hydrophily)

- Quite rare in flowering plants; limited to about **30 genera, mostly monocotyledons**. (By contrast, water is the regular mode of male gamete transport in lower plants like algae, bryophytes, and pteridophytes, restricting their geographical distribution.)
- **Crucial Exception (Aquatic Plants Pollinated by Insects or Wind):** Not all aquatic plants use water for pollination! In a majority of aquatic plants such as **water hyacinth (*Eichhornia*) and water lily (*Nymphaea*)**, the flowers emerge above the water level and are pollinated by **insects or wind**.
- **True Water-Pollinated Examples:**

- **Freshwater plants:** *Vallisneria* and *Hydrilla*. In ***Vallisneria*** **(epihydrophily)**, the female flower reaches the surface of water by a long coiled stalk, and male flowers or pollen grains are released onto the water surface and carried passively by water currents to the stigma.
- **Marine sea-grasses:** ***Zostera*** **(hypohydrophily)**. Female flowers remain **submerged in water**, and pollen grains are released inside the water. Pollen grains in such species are **long, ribbon-like** and carried passively inside the water.

- In most water-pollinated species, pollen grains are protected from wetting by a **mucilaginous covering**.

C. Animal Pollination (Zoophily)

- **Pollinators:** Bees, butterflies, flies, beetles, wasps, ants, moths, birds (**sunbirds and hummingbirds**), and bats. Among animals, **insects—particularly bees—are the dominant biotic pollinating agents**.
- **Unusual Vertebrate Pollinators:** Larger animals such as **primates (lemurs)**, **arboreal (tree-dwelling) rodents**, and **reptiles (gecko lizard and garden lizard)** also act as pollinators in some species.
- **Floral Adaptations & Rewards:**
  - Insect-pollinated flowers are generally **large, colourful, fragrant, and rich in nectar**. Small flowers cluster into a conspicuous **inflorescence**. Pollen grains are generally **sticky**.
  - Flowers pollinated by **flies and beetles secrete foul odours** to attract these insects.
  - **Usual Floral Rewards: Nectar and pollen grains**.
  - **Safe Place to Lay Eggs as a Floral Reward:**
    1. ***Amorphophallus*** (tallest flower, about 6 feet in height) provides a safe site for insects to lay eggs.
    2. ***Yucca*** **and Moth obligate mutualism:** Neither species can complete its life cycle without the other. The moth deposits its eggs in the **locule of the ovary** while pollinating the *Yucca* flower; the moth larvae hatch as the seeds begin developing.

  - **Pollen / Nectar Robbers:** Floral visitors that consume pollen or nectar **without bringing about pollination** (without contacting both anthers and stigma).

5.3 Outbreeding Devices

Continued self-pollination results in **inbreeding depression**. To discourage self-pollination and promote cross-pollination, flowering plants have evolved four major outbreeding devices:

1. **Non-synchronisation (Dichogamy):** Pollen release and stigma receptivity are not synchronised (either pollen is released before the stigma becomes receptive, or stigma becomes receptive much before pollen release). Prevents autogamy.
2. **Different positions of anther and stigma (Herkogamy):** Anther and stigma are placed at different positions so pollen cannot contact the stigma of the same flower. Prevents autogamy.
3. **Self-incompatibility:** A **genetic mechanism** that prevents self-pollen (from the same flower or other flowers of the same plant) from fertilising the ovules by **inhibiting pollen germination or pollen tube growth in the pistil**. Prevents both autogamy and geitonogamy.
4. **Production of unisexual flowers (Dicliny):**
   - **Monoecious plants** (both male and female flowers on the same plant, e.g., **castor and maize**): **Prevents autogamy, but does NOT prevent geitonogamy**.
   - **Dioecious plants** (male and female flowers on different plants, e.g., **papaya, date palm,** ***Vallisneria***): **Prevents both autogamy and geitonogamy**.

5.4 Pollen-Pistil Interaction & Artificial Hybridisation

Pollen-pistil interaction encompasses all events from **pollen deposition on the stigma until the pollen tube enters the ovule**.

- It is a dynamic process mediated by **chemical components** of the pollen interacting with those of the pistil, enabling the pistil to recognise compatible (right type) vs. incompatible (wrong type or self-incompatible) pollen.

- Following compatible pollination, the pollen grain germinates on the stigma to produce a pollen tube through one of the **germ pores**.
- If pollen was shed at the 2-celled stage, the **generative cell divides to form two male gametes during the growth of the pollen tube in the stigma**. (If shed at the 3-celled stage, the tube carries two male gametes from the beginning.)
- The pollen tube grows through the tissues of the stigma and style, enters the ovule through the **micropyle**, and then enters **one of the synergids through the filiform apparatus** (which chemically guides the pollen tube).
- **Laboratory Pollen Germination Observation:** Dusting pollen from flowers such as **pea, chickpea, *Crotalaria*, balsam, and *Vinca*** on a glass slide containing a drop of **about 10% sugar solution** allows observation of emerging pollen tubes under a low-power microscope within 15–30 minutes.

Artificial Hybridisation (Crop Improvement Technique)

To cross different species or genera and combine desirable characters without contamination by unwanted pollen, breeders use **emasculation and bagging**:

- **If the female parent bears bisexual flowers:**
  1. **Emasculation:** Removal of anthers from the flower bud **before the anther dehisces** using a pair of forceps.
  2. **Bagging:** Covering the emasculated flower with a bag (generally made of butter paper) to prevent contamination of its stigma with unwanted pollen.
  3. **Controlled Pollination & Rebagging:** When the bagged stigma attains receptivity, mature pollen from the desired male parent is dusted onto the stigma, and the flower is **rebagged** until fruits develop.
- **If the female parent produces unisexual flowers:** There is **no need for emasculation**. The female flower buds are simply **bagged before the flowers open**, pollinated when receptive, and rebagged.

6. Double Fertilisation

After entering one of the synergids through the filiform apparatus, the pollen tube releases the **two male gametes** into the **cytoplasm of the synergid**. Two fusion events then occur inside the embryo sac:

1. $\text{Syngamy (Generative Fertilisation): Male Gamete } (n) + \text{Egg Cell } (n) \longrightarrow \text{Zygote } (2n) \longrightarrow$
2. $\text{Triple Fusion (Vegetative Fertilisation): Male Gamete } (n) + \text{Two Polar Nuclei } (n+n) \longrightarrow$

- Because both **syngamy** and **triple fusion** take place in an embryo sac, the phenomenon is termed **double fertilisation**—an event **unique to flowering plants (angiosperms)**.
- The central cell after triple fusion becomes the **Primary Endosperm Cell (PEC)** and develops into the triploid $(3n)$ nutritive **endosperm**, while the diploid $(2n)$ **zygote** develops into the **embryo**.
- Meanwhile, the synergids and antipodal cells degenerate after fertilisation.

7. Post-Fertilisation: Structures and Events

Following double fertilisation, the events of **endosperm development, embryo development (embryogeny), maturation of ovule(s) into seed(s), and ovary into fruit** are collectively termed **post-fertilisation events**.

7.1 Endosperm Development

Endosperm development always precedes embryo development to ensure that nutritive tissue is available for the developing embryo. The Primary Endosperm Cell (PEC) divides repeatedly to form a **triploid $(3n)$ endosperm tissue** filled with reserve food materials.

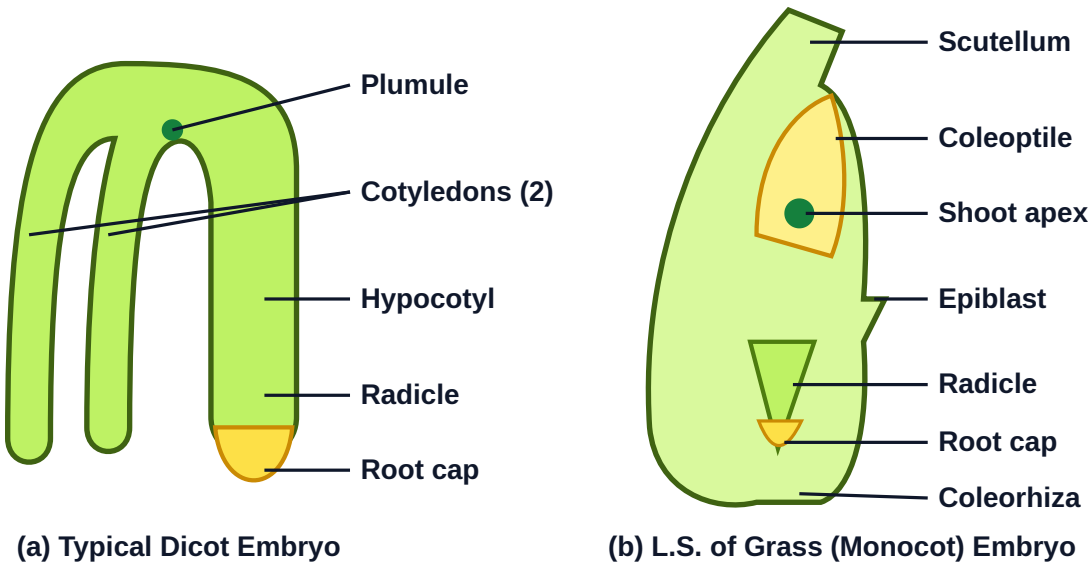
- **Free-Nuclear vs. Cellular Endosperm:** In the most common type of endosperm development, the Primary Endosperm Nucleus (PEN) undergoes successive mitotic nuclear divisions without cell wall formation to give rise to free nuclei (**free-nuclear endosperm**). Subsequently, cell wall formation occurs and the endosperm becomes **cellular**.
  - **Coconut Example:** The **coconut water** from tender coconut is **free-nuclear endosperm** (made up of thousands of nuclei), while the surrounding **white kernel** is the **cellular endosperm**.

- **Fate of Endosperm in Mature Seeds:**
  - **Completely consumed** by the developing embryo before seed maturation (**Non-albuminous / Ex-albuminous seeds**): **Pea, groundnut, beans**.
  - **Persists in the mature seed** and is used during seed germination (**Albuminous / Endospermic seeds**): **Castor, coconut, wheat, maize, barley, rice, sunflower**.

7.2 Embryo Development (Embryogeny)

The embryo develops at the **micropylar end** of the embryo sac where the zygote is situated. Most zygotes divide only after a certain amount of endosperm is formed. Early stages of embryogeny are similar in both monocotyledons and dicotyledons. In a dicotyledonous embryo, the zygote gives rise to:

$$\text{Zygote } (2n) \longrightarrow \text{Proembryo} \longrightarrow \text{Globular Embryo} \longrightarrow \text{Heart-Shaped Embryo} \longrightarrow \text{Mature E}$$



(a) Typical Dicot Embryo **(b) L.S. of Grass (Monocot) Embryo**

Tutorial Diagram 6: Structure of (a) a typical dicotyledonous embryo and (b) longitudinal section (L.S.) of a monocotyledonous grass embryo.

Feature	Dicotyledonous Embryo	Monocotyledonous (Grass) Embryo
Cotyledons	Possesses **two cotyledons** attached laterally to the embryonal axis.	Possesses only **one cotyledon**, called the **scutellum** in the grass family, situated towards one side (lateral) of the embryonal axis. (A rudimentary second cotyledon, the **epiblast**, may be present.)
Upper Axis (Epicotyl)	The portion of the embryonal axis **above the level of cotyledons** is the **epicotyl**, which terminates with the **plumule** (stem tip).	The epicotyl (above the level of attachment of the scutellum) has a shoot apex and a few leaf primordia enclosed in a hollow foliar structure called the **coleoptile**.
Lower Axis (Hypocotyl & Radicle)	The cylindrical portion **below the level of cotyledons** is the **hypocotyl**, terminating at its lower end in the **radicle** (root tip) covered by a **root cap**.	At its lower end, the embryonal axis has the radicle and root cap enclosed in an undifferentiated sheath called the **coleorhiza**.

7.3 Seed and Fruit Development

In angiosperms, the **seed is the final product of sexual reproduction** and is described as a **fertilised ovule**. A seed typically consists of **seed coat(s), cotyledon(s), and an embryo axis**.

- **Perisperm (Persistent Nucellus):** Occasionally, in some seeds such as **black pepper and beet**, remnants of the nucellus also persist in the mature seed. This residual, persistent nucellus ($2n$) is called **perisperm**.
- **Seed Coat & Micropyle:** Integuments of ovules harden into tough protective seed coats (outer **testa** and inner **tegmen**). The **micropyle** remains as a small pore in the seed coat, facilitating the **entry of oxygen and water** into the seed during germination.
- **Maturation, Dehydration & Dormancy:** As the seed matures, its water content decreases and seeds become relatively dry (**10–15% moisture by mass**). Metabolic activity slows down, and the embryo may enter a state of inactivity called **dormancy**, or germinate if favourable conditions (adequate moisture, oxygen, and suitable temperature) are present.Dehydration and dormancy are crucial for year-round seed storage in agriculture.
- **Record Seed Viability:**
  - ***Lupinus arcticus* (Arctic lupine):** Excavated from the **Arctic Tundra**; germinated and flowered after an estimated record of **10,000 years of dormancy** (oldest viable seed).
  - ***Phoenix dactylifera* (Date palm):** A **2,000-year-old** viable seed discovered during archaeological excavation at **King Herod's palace near the Dead Sea**.

- **Enormous Seed Production in Tiny-Seeded Plants:** Fruits of **orchids** and parasitic plant species such as ***Orobanche* and *Striga*** contain **thousands of tiny seeds** per fruit. Similarly, a tiny seed of ***Ficus*** develops into a massive tree producing billions of seeds.

Transformation of Ovary into Fruit

Simultaneously with ovules maturing into seeds, the **ovary develops into a fruit**, and the wall of the ovary develops into the fruit wall called the **pericarp**.

- **Fleshy fruits:** Guava, orange, mango.
- **Dry fruits:** Groundnut, mustard.
- **True Fruits:** Develop **only from the ovary** (floral parts degenerate and fall off).
- **False Fruits:** Fruits in which the **thalamus** (or other floral parts) also contributes to fruit formation. Examples: **Apple, strawberry, cashew**.
- **Parthenocarpic Fruits:** Fruits that develop **without fertilisation**. Example: **Banana**. Parthenocarpy can be induced through the application of growth hormones (auxins/gibberellins), and such fruits are **seedless**. (Note: One would NOT induce parthenocarpy in plants like pomegranate where the edible part is part of the seed/testa, or where seeds themselves are harvested!)

8. Apomixis and Polyembryony

8.1 Apomixis

Although seeds are generally products of fertilisation, a few flowering plants—such as some species of ***Asteraceae* and grasses**—have evolved a special mechanism to **produce seeds without fertilisation**, called **apomixis**.

- Apomixis is a form of **asexual reproduction that mimics sexual reproduction**.
- **Mechanisms of Apomictic Seed Development:**
  1. In some species, a **diploid egg cell ($2n$) is formed without reduction division (meiosis)** and develops into an embryo without fertilisation (recurrent agamospermy / diplospory).
  2. More often, as in many ***Citrus* and *Mango*** varieties, some of the diploid **nucellar cells ($2n$) surrounding the embryo sac start dividing, protrude into the embryo sac, and develop into embryos** (adventive embryony).

- Because apomictic embryos develop mitotically from maternal diploid cells without meiosis or syngamy, they are genetically identical to the parent plant and can be called **clones**.

8.2 Polyembryony

The occurrence of **more than one embryo in a seed** is referred to as **polyembryony** (first observed in *Citrus*/orange seeds). Squeezing a seed of orange reveals multiple embryos of varying sizes and shapes.

Importance of Apomixis in the Hybrid Seed Industry

- **The Problem with Hybrid Seeds:** Hybrid varieties of food and vegetable crops tremendously increase productivity, but **hybrid seeds must be produced every year**. If seeds collected from hybrid plants are sown, the progeny plants **segregate** (due to meiosis and recombination) and do not maintain hybrid characters. Producing hybrid seeds annually by hand-pollination is expensive for farmers.
- **The Apomictic Solution:** If hybrids are engineered into **apomicts**, there is **no segregation of characters** in the hybrid progeny. Farmers can use their own harvested hybrid seeds to raise new crops year after year without buying new hybrid seeds annually.

9. Key Conceptual Comparisons (NCERT Exercises & High-Yield Tables)

Pair of Terms	Structure / Concept 1	Structure / Concept 2
Microsporogenesis vs. Megasporogenesis	Formation of haploid microspores (n) from a diploid PMC ($2n$) via meiosis inside the microsporangium (anther). All 4 microspores in a tetrad are functional.	Formation of haploid megaspores (n) from a diploid MMC ($2n$) via meiosis inside the megasporangium (ovule). Usually 3 degenerate and only 1 functional megaspore forms the embryo sac.
Hypocotyl vs. Epicotyl	Cylindrical portion of the embryonal axis **below the level of cotyledons** that terminates in the radicle (root tip).	Portion of the embryonal axis **above the level of cotyledons** that terminates with the plumule (stem tip).
Coleoptile vs. Coleorhiza	Hollow foliar protective sheath enclosing the **shoot apex and leaf primordia** in a monocot (grass) embryo.	Undifferentiated protective sheath enclosing the **radicle and root cap** at the lower end of a monocot (grass) embryonal axis.
Integument vs. Testa	Pre-fertilisation protective envelope(s) encircling the nucellus of an ovule.	Post-fertilisation hardened outer protective **seed coat** formed from the outer integument of the ovule.
Perisperm vs. Pericarp	Residual, persistent **nucellus** ($2n$) inside a mature seed (e.g., black pepper, beet).	The wall of the **fruit** ($2n$) that develops from the wall of the ovary after fertilisation.
Parthenocarpy vs. Apomixis	Formation of a **seedless fruit** without fertilisation (e.g., banana).	Formation of viable **seeds** without fertilisation (e.g., *Asteraceae*, grasses, *Citrus*).

10. Worked Examples (Quantitative & Ploidy Problems for NEET UG)

Worked Example 1: Meiotic Divisions Required for Seed Formation

Problem: Calculate the total number of meiotic divisions required to produce 100 viable wheat grains (seeds), assuming all functional gametes participate successfully in fertilisation.

Step-by-Step Solution:

1. Each wheat grain (seed) requires the fertilisation of 1 female gametophyte (embryo sac) by 1 male gametophyte (pollen grain).
2. **In the Male (Microsporogenesis):** 1 meiotic division of a Pollen Mother Cell (PMC) produces 4 functional microspores (pollen grains). Therefore, the number of meiotic divisions needed to produce N pollen grains is $\frac{N}{4}$. For $N = 100$:
$$\text{Male meiotic divisions} = \frac{100}{4} = 25$$
3. **In the Female (Megasporogenesis):** 1 meiotic division of a Megaspore Mother Cell (MMC) produces 4 megaspores, out of which 3 degenerate and only 1 functional megaspore forms the embryo sac (monosporic development). Therefore, the number of meiotic divisions needed to produce N functional embryo sacs is N. For $N = 100$:
$$\text{Female meiotic divisions} = 100$$
4. **Total Meiotic Divisions:**
$$\text{Total Meiotic Divisions} = N + \frac{N}{4} = \frac{5N}{4} = 100 + 25 = 125$$

Answer: 125 meiotic divisions.

Worked Example 2: Asymmetric Ploidy Cross Calculation

Problem: A tetraploid female angiosperm plant ($4n = 48$) is crossed with a diploid male plant ($2n = 24$). Determine the chromosome number in the (a) nucellus, (b) zygote, (c) endosperm, and (d) antipodal cells of the resulting fertilised ovule.

Step-by-Step Solution:

1. **Haploid gametic sets (n):**
   - Female parent somatic chromosome number $= 4n = 48 \implies$ Female gamete (egg cell, polar nucleus, antipodal cell) receives half the maternal chromosomes via meiosis $= 2n = 24$ chromosomes.
   - Male parent somatic chromosome number $= 2n = 24 \implies$ Male gamete receives half the paternal chromosomes via meiosis $= n = 12$ chromosomes.
2. **(a) Nucellus:** Purely maternal sporophytic tissue $= 4n = 48$ chromosomes.
3. **(b) Zygote:** Formed by syngamy of 1 female egg cell $(24) + 1$ male gamete $(12) = 24 + 12 = 36$ chromosomes (triploid, $3n$).
4. **(c) Endosperm (from PEN):** Formed by triple fusion of 2 female polar nuclei $(24 + 24) + 1$ male gamete $(12) = 24 + 24 + 12 = 60$ chromosomes (pentaploid, $5n$).
5. **(d) Antipodal cells:** Unfertilised cells of the female gametophyte $= 24$ chromosomes.

Answer: Nucellus $= 48$; Zygote $= 36$; Endosperm $= 60$; Antipodal cells $= 24$.

11. NEET UG Practice Questions with Complete Worked Solutions

Note: All questions below are original, exam-aligned practice questions designed strictly according to the NEET UG standard and the supplied chapter content.

Practice Question 1 (Single-Choice MCQ — Anther Anatomy & Microsporogenesis)

Arrange the following structures and developmental stages of the male reproductive unit in the correct sequential order from earliest precursor to final functional structure:

I. Microspore tetrad • II. Sporogenous tissue • III. Male gametes • IV. Pollen grain • V. Pollen mother cell (PMC)

(A) II $\rightarrow$ V $\rightarrow$ I $\rightarrow$ IV $\rightarrow$ III

(B) V $\rightarrow$ II $\rightarrow$ I $\rightarrow$ IV $\rightarrow$ III

(C) II $\rightarrow$ I $\rightarrow$ V $\rightarrow$ IV $\rightarrow$ III

(D) II $\rightarrow$ V $\rightarrow$ IV $\rightarrow$ I $\rightarrow$ III

Correct Answer: (A)

Worked Solution: In a young anther, the homogeneous **sporogenous tissue (II)** occupies the centre of each microsporangium. Its cells differentiate into potential **pollen mother cells (V)**, which undergo meiosis (microsporogenesis) to form a **microspore tetrad (I)**. Upon anther dehydration, the microspores dissociate and mature into **pollen grains (IV)**, inside which the generative cell divides mitotically to form two **male gametes (III)**.

Practice Question 2 (Multiple-Statement Evaluation — Pollen Grain & Sporopollenin)

Read the following four statements (I–IV) regarding angiosperm pollen grains and select the option that identifies all correct statements:

I. The outer wall layer (exine) is continuous and composed of cellulose and pectin, whereas the inner layer (intine) is composed of sporopollenin.

II. In over 60 per cent of angiosperms, pollen grains are shed at the 2-celled stage consisting of a large vegetative cell and a small spindle-shaped generative cell.

III. Pollen grains of rice and wheat retain their viability for several months after release, whereas those of Rosaceae lose viability within 30 minutes.

IV. Pollen grains can be cryopreserved for years in liquid nitrogen at $-196\ ^{\circ}\mathrm{C}$ for use in pollen banks.

(A) I, II, and IV only

(B) II and IV only

(C) II, III, and IV only

(D) I and III only

Correct Answer: (B)

Worked Solution: Statements II and IV are correct. Statement I is incorrect because the **exine** is made of **sporopollenin** and is discontinuous at germ pores, whereas the **intine** is thin, continuous, and made of **cellulose and pectin**. Statement III is inverted: cereals like **rice and wheat lose pollen viability within 30 minutes**, while members of **Rosaceae, Leguminosae, and Solanaceae** retain viability for months.

Practice Question 3 (Assertion–Reason — Tapetum Characteristics)

Assertion (A): Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus.
Reason (R): The tapetum is the innermost wall layer of the microsporangium that nourishes the developing pollen grains and undergoes nuclear divisions without immediate cytokinesis.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Solution: The tapetum is metabolically hyperactive because its primary role is to nourish developing microspores and synthesize sporopollenin precursors. High metabolic demand and free nuclear division (or endomitosis without cell wall formation) explain why tapetal cells have dense cytoplasm and become binucleate or multinucleate.

Practice Question 4 (Single-Choice MCQ — Gynoecium & Ovule Number)

Which of the following correctly pairs the gynoecium/ovary characteristic with its accurate botanical example(s)?

(A) Multicarpellary, apocarpous gynoecium — *Papaver*

(B) Multicarpellary, syncarpous gynoecium — *Michelia*

(C) Single ovule in each ovary — Wheat, paddy, and mango

(D) Single ovule in each ovary — Papaya, watermelon, and orchids

Correct Answer: (C)

Worked Solution: Wheat, paddy (rice), and mango have a single ovule in each ovary. In contrast, papaya, watermelon, and orchids have many ovules per ovary. *Papaver* has a multicarpellary **syncarpous** pistil, whereas *Michelia* has a multicarpellary **apocarpous** gynoecium.

Practice Question 5 (Single-Choice MCQ — Monosporic Embryo Sac Development)

How many meiotic divisions and sequential mitotic nuclear divisions, respectively, are required to form a mature 7-celled, 8-nucleate embryo sac starting from a single megaspore mother cell (MMC)?

(A) 1 meiotic division and 2 mitotic divisions

(B) 1 meiotic division and 3 mitotic divisions

(C) 2 meiotic divisions and 3 mitotic divisions

(D) 3 meiotic divisions and 1 mitotic division

Correct Answer: (B)

Worked Solution: A single diploid MMC ($2n$) undergoes **1 complete meiotic division** (Meiosis I and II) to produce 4 haploid megaspores, of which 3 degenerate. The single functional megaspore nucleus then undergoes **3 sequential free-nuclear mitotic divisions** ($1 \rightarrow 2 \rightarrow 4 \rightarrow 8$ nuclei) to form the 8-nucleate, 7-celled mature embryo sac.

Practice Question 6 (Matching Type — Pollination Ecology & Examples)

Match Column I (Plant Species) with Column II (Pollination Mechanism / Feature):

Column I (Plant)	Column II (Feature / Mechanism)
P. *Vallisneria*	1. Aquatic plant emerging above water surface; pollinated by insects or wind
Q. *Zostera*	2. Produces both chasmogamous and underground cleistogamous flowers
R. Water hyacinth & Water lily	3. Submerged marine seagrass with long, ribbon-like pollen grains
S. *Commelina*	4. Freshwater plant where female flowers reach the water surface via a long stalk

(A) P–4, Q–3, R–1, S–2

(B) P–3, Q–4, R–1, S–2

(C) P–4, Q–1, R–3, S–2

(D) P–2, Q–3, R–4, S–1

Correct Answer: (A)

Worked Solution: *Vallisneria* (P–4) exhibits surface water pollination in fresh water. *Zostera* (Q–3) is a marine seagrass exhibiting underwater pollination with ribbon-like pollen. Water hyacinth and water lily (R–1) are aquatic plants whose flowers emerge above water and are pollinated by insects or wind. *Commelina* (S–2), along with *Viola* and *Oxalis*, produces both open chasmogamous and closed cleistogamous flowers.

Practice Question 7 (Single-Choice MCQ — Outbreeding Devices)

Compare the reproductive strategies of castor/maize with that of papaya:

(A) Both castor and papaya prevent both autogamy and geitonogamy.

(B) Castor and maize (monoecious) prevent autogamy but not geitonogamy, whereas papaya (dioecious) prevents both autogamy and geitonogamy.

(C) Castor and maize prevent geitonogamy but not autogamy, whereas papaya prevents only autogamy.

(D) Both castor and papaya produce cleistogamous flowers to ensure self-pollination.

Correct Answer: (B)

Worked Solution: In monoecious plants like **castor and maize**, unisexual male and female flowers are borne on the same plant; this prevents autogamy (since individual flowers are unisexual) but allows pollen transfer between different flowers of the same plant (**geitonogamy**). In dioecious plants like **papaya**, male and female flowers are on separate plants, preventing **both autogamy and geitonogamy**.

Practice Question 8 (Assertion–Reason — Artificial Hybridisation)

Assertion (A): When the female parent selected for an artificial hybridisation programme bears unisexual flowers, emasculation is not required.
Reason (R): Unisexual female flowers lack stamens (anthers), so there is no risk of self-pollen shedding within the flower bud prior to bagging.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Solution: Emasculation is the surgical removal of anthers from a bisexual flower bud before dehiscence to prevent self-pollination. Because unisexual female flowers have no anthers, emasculation is unnecessary; the female flower buds only need to be bagged before opening.

Practice Question 9 (Quantitative Ploidy Problem — Double Fertilisation)

In a diploid angiosperm species, the cells of the root tip contain $2n = 16$ chromosomes. What will be the number of chromosomes in the (i) synergid, (ii) primary endosperm nucleus (PEN), (iii) coleoptile cell, and (iv) perisperm cell (if present), respectively?

(A) 8, 24, 16, 24

(B) 8, 24, 16, 16

(C) 16, 24, 8, 16

(D) 8, 16, 16, 24

Correct Answer: (B)

Worked Solution: Given $2n = 16$, the haploid number is $n = 8$.
(i) **Synergid:** Haploid cell of female gametophyte $= n = 8$.
(ii) **Primary Endosperm Nucleus (PEN):** Triploid product of triple fusion $= 3n = 3 \times 8 = 24$.
(iii) **Coleoptile cell:** Diploid part of the mature embryo $= 2n = 16$.
(iv) **Perisperm cell:** Persistent maternal nucellus $= 2n = 16$.

Practice Question 10 (Single-Choice MCQ — Seed Classification & Perisperm)

Identify the correct set of examples for (I) Non-albuminous seeds, (II) Albuminous seeds, and (III) Perispermic seeds:

(A) I: Pea, groundnut; II: Castor, maize, wheat; III: Black pepper, beet

(B) I: Castor, maize; II: Pea, groundnut; III: Apple, strawberry

(C) I: Pea, groundnut; II: Black pepper, beet; III: Wheat, barley

(D) I: Wheat, barley; II: Pea, beans; III: Black pepper, castor

Correct Answer: (A)

Worked Solution: Non-albuminous (ex-albuminous) seeds have no residual endosperm at maturity because the embryo consumes it completely (**pea, groundnut, beans**). Albuminous seeds retain residual endosperm (**wheat, maize, barley, castor, coconut**). Seeds with persistent nucellus (**perisperm**) include **black pepper and beet**.

Practice Question 11 (Multiple-Statement Evaluation — Fruits, Seed Viability & Apomixis)

Consider the following statements (I–IV):

I. In apple, strawberry, and cashew, the thalamus contributes to fruit formation, making them false fruits.

II. A 10,000-year-old viable seed of *Phoenix dactylifera* was excavated from the Arctic Tundra, whereas a 2,000-year-old viable seed of *Lupinus arcticus* was found near the Dead Sea.

III. In *Citrus* and *Mango* varieties, diploid nucellar cells surrounding the embryo sac protrude into the embryo sac and develop into multiple embryos, resulting in polyembryony.

IV. Introducing apomictic genes into hybrid crop varieties prevents the segregation of characters in the hybrid progeny.

Which of the above statements are correct?

(A) I, II, and III only

(B) I, III, and IV only

(C) II and IV only

(D) I, II, III, and IV

Correct Answer: (B)

Worked Solution: Statements I, III, and IV are completely accurate. Statement II swaps the two historical seed records: ***Lupinus arcticus*** (lupine) was excavated from the **Arctic Tundra (10,000 years of dormancy)**, whereas ***Phoenix dactylifera*** (date palm) was discovered at **King Herod's palace near the Dead Sea (2,000 years of dormancy)**.

Practice Question 12 (Single-Choice MCQ — Pollen-Pistil Interaction & Filiform Apparatus)

During pollen-pistil interaction in a species that sheds pollen at the 2-celled stage, where does the mitotic division of the generative cell occur, and where are the two male gametes discharged?

(A) Generative cell divides inside the microsporangium; male gametes are discharged directly into the central cell.

(B) Generative cell divides during the growth of the pollen tube in the stigma; male gametes are discharged into the cytoplasm of one of the synergids.

(C) Generative cell divides inside the ovary locule; male gametes are discharged into the antipodal cells.

(D) Generative cell divides after entering the synergid; one gamete is discharged in the micropyle and the other in the funicle.

Correct Answer: (B)

Worked Solution: In plants that shed pollen at the 2-celled condition, the generative cell divides mitotically to form the two male gametes **during the growth of the pollen tube in the stigma**. After entering the ovule through the micropyle, the pollen tube is guided by the filiform apparatus into **one of the synergids**, where it discharges both male gametes into the **synergid's cytoplasm**.

12. Common Exam Pitfalls & Revision Checklist

High-Frequency NEET UG Traps to Avoid

1. **Aquatic Plant Pollination Trap:** Never assume all aquatic plants are water-pollinated! **Water hyacinth (*Eichhornia*) and water lily (*Nymphaea*)** bear flowers above the water surface and are pollinated by **insects or wind**. Only ~30 genera (mostly monocots like *Vallisneria*, *Hydrilla*, and marine *Zostera*) use water pollination.
2. **Geitonogamy Functional vs. Genetic Nature:** Geitonogamy is **functionally cross-pollination** (requires a pollinating agent) but **genetically self-pollination (autogamy)** because the pollen originates from the same plant.
3. **Monoecious vs. Dioecious Outbreeding Control:** Monoecious plants (**castor, maize**) prevent autogamy ONLY; they do NOT prevent geitonogamy. Dioecious plants (**papaya**) prevent BOTH autogamy and geitonogamy.
4. **Perisperm vs. Endosperm Ploidy:** Perisperm (in **black pepper and beet**) is persistent maternal **nucellus** and is **diploid ($2n$)**, whereas angiosperm endosperm is formed by triple fusion and is **triploid ($3n$)**.
5. **Orientation of Embryo Sac Diagrams:** Textbook diagrams show the embryo sac either with the micropylar end at the top or at the bottom. Always identify the **micropylar end by the filiform apparatus / synergids / egg cell**, and the **chalazal end by the three antipodal cells**.
6. **Coconut Water vs. White Kernel:** Tender coconut water is **free-nuclear endosperm** (thousands of free triploid nuclei), whereas the surrounding white edible kernel is **cellular endosperm**.

13. Concise Final Recap

- **Male Reproductive Unit:** Stamen $=$ Filament $+$ Bilobed, dithecous, tetrasporangiate Anther. Microsporangium wall layers: Epidermis, Endothecium, Middle layers (protection & dehiscence), and multinucleate Tapetum (nutrition). PMC $(2n) \xrightarrow{\text{Meiosis}}$ 4 haploid microspores $(n) \rightarrow$ Pollen grains ($25\text{–}50\ \mu\text{m}$; sporopollenin exine with germ pores + cellulose-pectin intine; shed at 2-celled stage in $> 60\%$ angiosperms).
- **Female Reproductive Unit:** Pistil $=$ Stigma $+$ Style $+$ Ovary containing ovules (megasporangia) attached to placenta via funicle at hilum. Single MMC $(2n)$ at micropylar nucellus $\xrightarrow{\text{Meiosis}}$ 4 megaspores (3 degenerate, 1 functional) $\xrightarrow{\text{3 free-nuclear mitoses}}$ **7-celled, 8-nucleate embryo sac** (2 synergids with filiform apparatus + 1 egg cell + 1 central cell with 2 polar nuclei + 3 antipodals).
- **Double Fertilisation:** Unique to angiosperms. **Syngamy** [Male gamete (n) + Egg $(n) \rightarrow$ Zygote $(2n)$] $+$ **Triple fusion** [Male gamete (n) + 2 Polar nuclei $(n + n) \rightarrow$ PEN $(3n)$].
- **Post-Fertilisation & Special Modes:** PEC $(3n)$ forms nutritive endosperm prior to embryogeny (Proembryo $\rightarrow$ Globular $\rightarrow$ Heart-shaped $\rightarrow$ Mature embryo). Ovule matures into seed (10–15% moisture); ovary matures into fruit (wall $=$ pericarp). **Apomixis** (*Asteraceae*, grasses) forms seeds without fertilisation; **Polyembryony** (*Citrus*, *Mango*) forms multiple embryos per seed via nucellar adventive embryony.